# EXPERIMENT 5
# TRANSIENT COUETTE FLOW

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**Aim:** To compare explicit FTCS and implicit backward-Euler numerical solutions for transient momentum diffusion in Couette flow with the analytical series solution.

## 1. Introduction

Transient Couette flow is the motion of a viscous fluid between two parallel plates when the upper plate moves suddenly at a constant speed while the lower plate remains stationary. The fluid initially has zero velocity. Momentum is transferred from the moving plate into the fluid by viscous diffusion. This experiment isolates one-dimensional viscous diffusion; pressure gradients and convective transport are not included.

## 2. Governing Equation and Boundary Conditions

For incompressible, one-dimensional flow with velocity u in the x-direction and variation only in the y-direction, the governing momentum-diffusion equation is:

$$\partial u/\partial t = \nu \, \partial^2 u/\partial y^2$$

Initial condition:

$$u(y, 0) = 0 \qquad \text{for } 0 \le y < H$$

Boundary conditions:

$$u(0, t) = 0 \qquad \text{and} \qquad u(H, t) = U$$

Here, H is the distance between the plates, U is the speed of the moving upper plate, and $\nu$ is the kinematic viscosity. The upper-plate velocity is imposed at the top boundary.

## 3. Numerical Methods

### 3.1 Explicit FTCS method

The Forward-Time Central-Space (FTCS) method advances the velocity explicitly using the previous time level:

$$u_i^{n+1} = u_i^n + Fo\,(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

The Fourier number is $Fo = \nu\Delta t/(\Delta y)^2$. For this one-dimensional diffusion equation, the FTCS stability condition is $Fo \leq 0.5$. The time step must therefore remain sufficiently small as the grid is refined.

### 3.2 Implicit backward-Euler method

The backward-Euler method evaluates the diffusion term at the new time level. This produces a tridiagonal system of algebraic equations at every time step, which is solved using the Thomas algorithm (TDMA) in the supplied solver. The method is unconditionally stable for the linear diffusion problem considered here, although a large time step can still reduce temporal accuracy.

## 4. Simulation Parameters and Grid

| Parameter | Value |
|---|---|
| Number of grid points in y (ny) | 81 |
| Plate spacing, H | 1.0 |
| Upper plate speed, U | 1.0 |
| Kinematic viscosity, ν | 0.01 |
| Final physical time, tEnd | 2.0 |
| Reynolds number, Re = UH/ν | 100 |
| Explicit target Fourier number | 0.45 |
| Explicit actual time step, Δt | 0.007017543859649 |
| Explicit actual Fourier number | 0.4491228070175 |
| Explicit number of time steps | 285 |
| Implicit target Fourier number | 5.0 |
| Implicit actual time step, Δt | 0.07692307692308 |
| Implicit actual Fourier number | 4.923076923077 |
| Implicit number of time steps | 26 |
| Output/history interval | Every 50 explicit steps; first and final steps also printed |

Grid spacing: $\Delta y = H/(ny - 1) = 1/80 = 0.0125$. The executed solver command used ny = 81. Although the notebook form variable is shown as ny = 80, the lab argument list and the recorded run command determine the actual simulation grid used for these results.

## 5. Stability Boundary

For explicit FTCS, the stability limit is $Fo \leq 0.5$. With $\Delta y = 0.0125$ and $\nu = 0.01$, the maximum permitted time step at the stability boundary is:

$$\Delta t_{max} = 0.5(\Delta y)^2/\nu = 0.0078125$$

The actual explicit time step was 0.00701754, giving Fo = 0.44912. This is below 0.5, so the explicit calculation satisfies the diffusion stability criterion. The implicit calculation used Fo = 4.92308, well above 0.5; it remains stable because backward Euler is implicit and unconditionally stable for this linear diffusion equation. Stability does not mean that a large time step is equally accurate.

## 6. Results and Quantitative Comparison

Both methods were compared with the analytical transient Couette-flow series solution at the same final physical time, t = 2.0. The relative L2 error is the norm of the difference between the numerical and analytical velocity profiles divided by the norm of the analytical profile.

| Comparison item | Explicit FTCS | Implicit backward Euler |
|---|---|---|
| Final physical time | 2.0 | 2.0 |
| Actual time step | 0.00701754 | 0.07692308 |
| Actual Fourier number | 0.44912281 | 4.92307692 |
| Number of time steps | 285 | 26 |
| Final relative L2 error | $6.110517 \times 10^{-4}$ | $6.993319 \times 10^{-3}$ |
| Stability | Stable ($Fo \leq 0.5$) | Stable (backward Euler) |

The explicit method produced the smaller final error: 0.000611052, compared with 0.00699332 for the implicit method. The implicit error is approximately 11.44 times the explicit error. However, the implicit method required only 26 time steps, compared with 285 for FTCS. Thus, it used about 10.96 times fewer time steps. This is a comparison of step counts, not a measured wall-clock runtime; each implicit step also requires solving a tridiagonal system.

## 7. Error History

The explicit relative L2 error decreased from $1.550244 \times 10^{-1}$ at its first step to $6.110517 \times 10^{-4}$ at t = 2.0. The implicit relative L2 error decreased from $1.296487 \times 10^{-1}$ at its first step to $6.993319 \times 10^{-3}$ at t = 2.0. The plotted history shows a decreasing error trend for both methods. The explicit curve lies below the implicit curve over the displayed time range, indicating closer agreement with the analytical solution for this selected set of time steps.

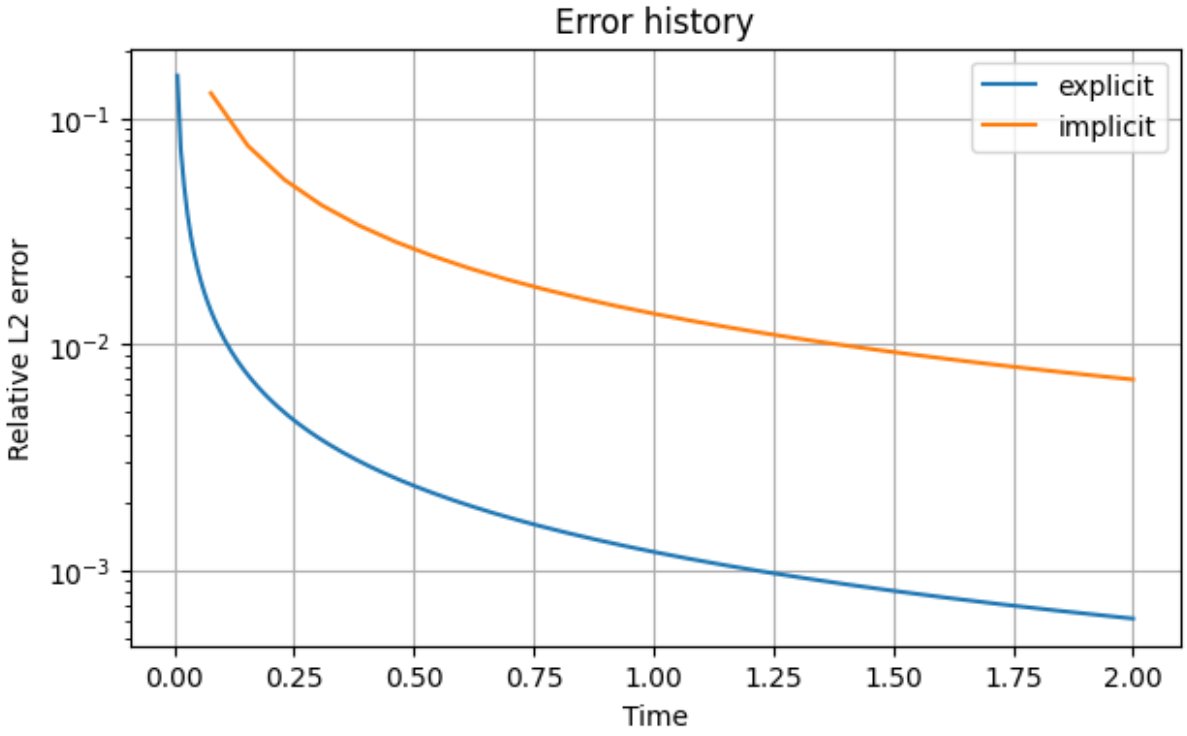



*Figure 1. Relative L2 error history for explicit FTCS and implicit backward-Euler solutions.*

## 8. Velocity Profile Comparison

At t = 2.0, the velocity is zero at the stationary lower plate and increases toward the upper-plate speed U = 1.0. The explicit, implicit, and analytical profiles are very close in the plotted figure. The explicit profile follows the analytical curve more closely, which is consistent with its smaller reported relative L2 error. The profile represents momentum diffusing upward from the moving plate into the initially stationary fluid.

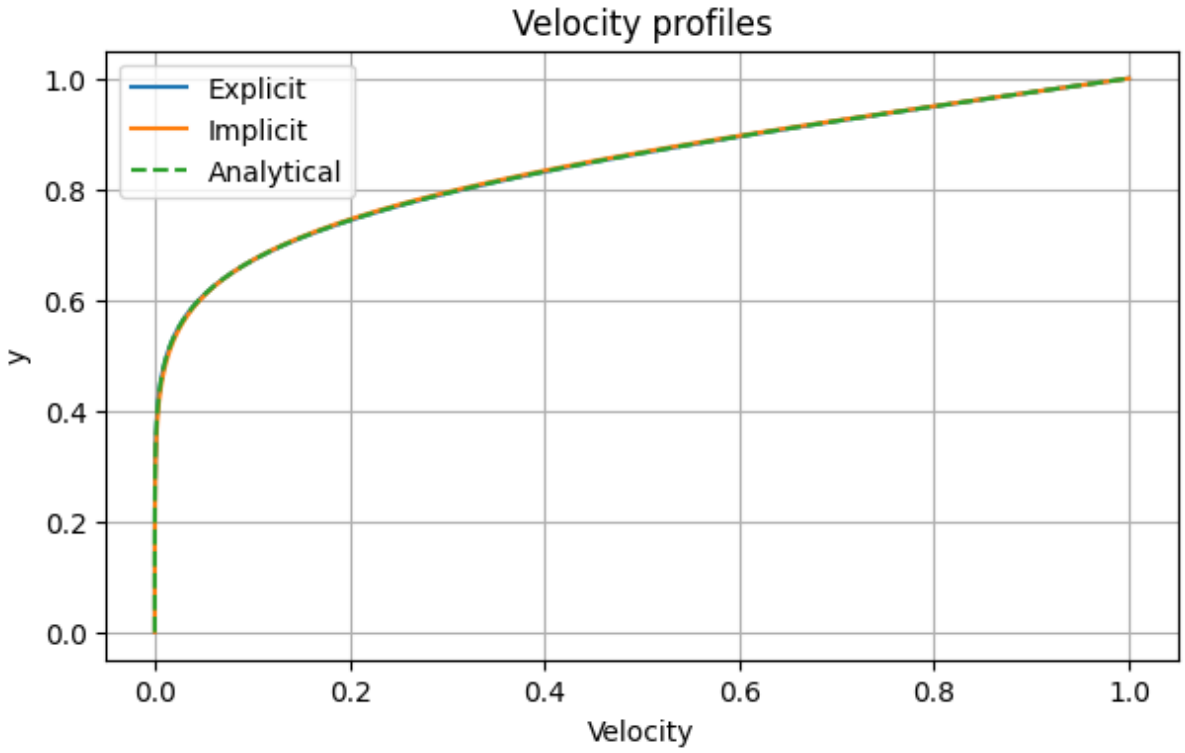



*Figure 2. Final velocity profiles from explicit FTCS, implicit backward Euler, and the analytical solution.*

## 9. Inline Velocity Field

The inline contour shows the explicit velocity field across the plate gap at the final time. The field is uniform in the x-direction, as expected for the one-dimensional Couette-flow model. Velocity is close to zero near the stationary lower plate and approaches the moving-plate speed near y = H. The colour gradient visualizes the vertical diffusion of momentum.

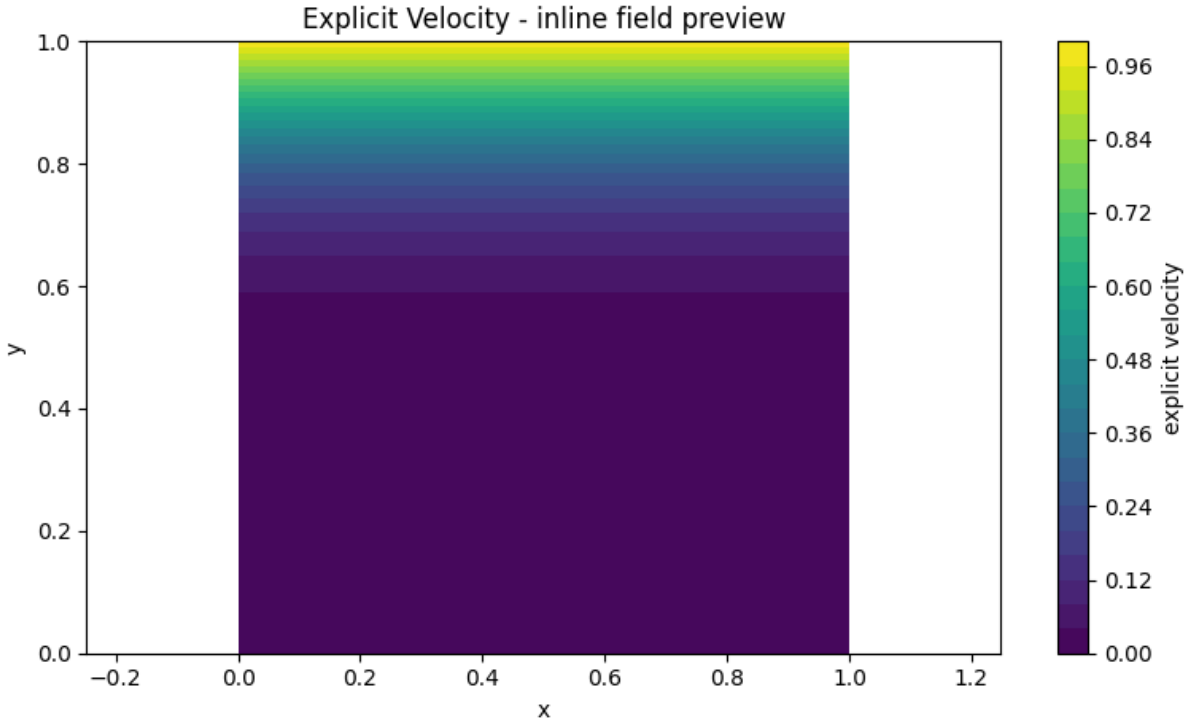



*Figure 3. Explicit velocity field at the final time; axes are x and y, and the colour bar represents velocity.*

## 10. Observed Numerical Behaviour and Physical Interpretation

### Observed numerical behaviour

- Both methods remained finite and produced decreasing relative L2 error histories for the reported run.
- The explicit FTCS method used a Fourier number below the 0.5 stability boundary and required 285 time steps.
- The implicit backward-Euler method used a much larger Fourier number and required only 26 time steps.

- At the same final time, the explicit method had a lower relative L2 error than the implicit method.
- The three final velocity profiles nearly overlap visually, although the quantitative errors distinguish their accuracy.

**Physical interpretation**

The moving upper plate transfers momentum to adjacent fluid layers through viscosity. This momentum diffuses toward the stationary lower plate with time. The resulting velocity increases from zero at the lower wall to the imposed speed at the upper wall. The plotted field and profile are consistent with this expected diffusion-dominated behaviour.

## 11. Limitations of the Teaching Model

- The model is one-dimensional and assumes velocity varies only across the gap between the plates.
- Pressure gradients, convection, and two-dimensional or three-dimensional effects are excluded.
- The fluid is treated as Newtonian with constant kinematic viscosity and constant properties.
- The plates are assumed parallel, with ideal no-slip boundary conditions and a prescribed constant upper-plate speed.
- The initial fluid velocity is taken as zero; start-up details beyond this idealized condition are not represented.
- The numerical solution is limited by the selected grid and time steps. The implicit method's unconditional stability does not remove time-discretization error.
- The comparison uses the analytical series solution implemented by the supplied teaching solver as its reference; it is not an experimental measurement.
- The provided output contains no wall-clock timing, mesh-refinement study, or formal order-of-accuracy study.

## 13. Conclusion

The transient Couette-flow simulation successfully compared explicit FTCS and implicit backward-Euler methods against the analytical solution at t = 2.0. The explicit method satisfied the stability requirement with Fo = 0.44912 and achieved a final relative L2 error of $6.110517 \times 10^{-4}$ using 285 steps. The implicit method remained stable at Fo = 4.92308 and required only 26 steps, but its final relative L2 error was higher at $6.993319 \times 10^{-3}$. Therefore, for this particular grid and time-step selection, explicit FTCS was more accurate, while backward Euler reduced the number of time steps substantially. The result demonstrates the trade-off between stability, accuracy, and computational effort in transient diffusion.